



GRAVITY
ELEVATING SOLUTIONS

ABOUT US

Gravity was founded in South Africa in 1999. Since then, Gravity has been a leader in the work-at-height industry, assisting in the development of a variety of standards, providing excellent training and producing innovative solutions for worksites and a range of fall arrest equipment.

Gravity is committed to providing quality services and products, with the clients' satisfaction always being the highest priority.

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Introduction

Gravity is a global leader in permanent fall protection for a range of applications and scenarios. Gravity has partnered with Fallprotec to manufacture rope and rail permanent lifelines.

Permanent systems require training specialised in installation and use.



Purpose

This booklet is a training aid that, together with the guidance of a competent facilitator, allows the learner to understand the requirements, limitations and safe use of Gravity Lifeline Systems (GLS).

Scope

Only the safe use of GLS will be discussed in this manual. There are two types of GLS:

- Gravity Vertical System (GVS)
- Gravity Horizontal System (GHS)

This manual does **not** extend to installations or detailed inspections of any systems, nor does it include the use of other permanent fall protection systems.

Should you encounter other Gravity systems, ensure that you receive the relevant training and follow the manufacturer's requirements.

Site Establishment

Before work starts:

- **Compliance** – Ensure that you comply with the client requirements onsite. For example, do not work at night if night work is not allowed.
- **Assistance** – Ensure that there are at least two workers onsite with the ability to rescue each other. This ensures that, in case of an emergency, help can be contacted and a rescue can be performed quickly.
- **Rescue** – Ensure that a rescue plan is in place and a sufficient rescue kit with a first aid kit is onsite.
- **Equipment** – Ensure that all equipment is inspected before use and is safe for use and fit for purpose.
- **Structure/Site** – Ensure that the structure/site has been inspected and deemed safe for use.
- **Traffic** – Ensure that a traffic management plan is in place (should traffic be a risk).
- **Assessment** – Ensure that a comprehensive risk assessment has been done and that all workers have been inducted on the risk assessment.
- **Relevant** – Ensure that the fall protection system is relevant and fit for purpose as per site and task.

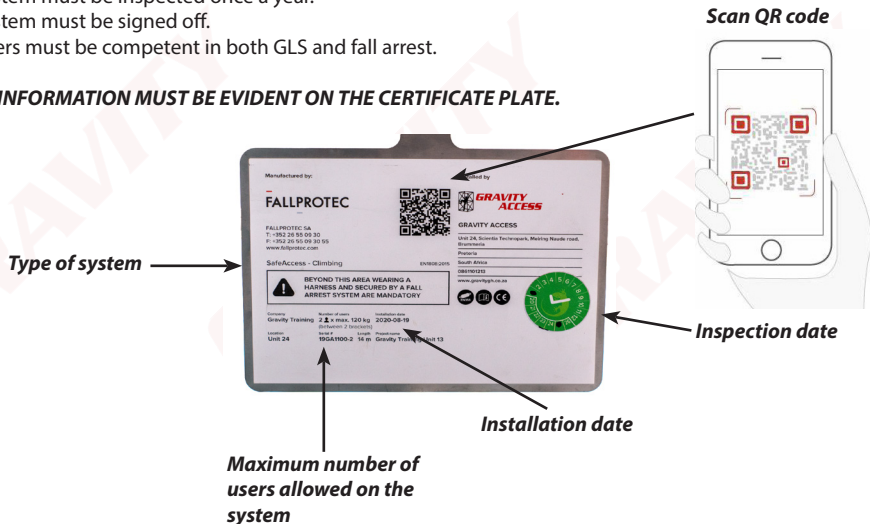


Requirements

Before any Gravity system may be used, the site supervisor must ensure that the following is in place:

- The system must be installed by a competent person.
- The system must be inspected once a year.
- The system must be signed off.
- The users must be competent in both GLS and fall arrest.

ALL OF THE INFORMATION MUST BE EVIDENT ON THE CERTIFICATE PLATE.



A. GRAVITY VERTICAL SYSTEM

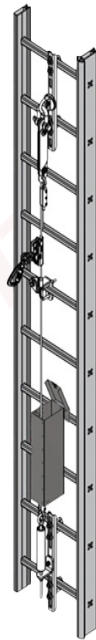
The GVS is constructed with an 8 mm 7x7 stainless steel wire rope running through intermediate anchors. The GVS is connected to end anchors at both ends, usually with a tensioning device at the bottom of the line. The GVS is complemented with a storage box for the glider/"ZIP!" fall arresters.

1. *Advantages and Disadvantages*

This system offers some advantages over other fall protection systems. It reduces the fall factor that workers are exposed to. It also reduces the severity of a potential fall. Furthermore, it allows for more efficient work onsite as it enables faster ascent and descent than would be possible with shock absorbing lanyards. Some systems also allow multiple glider/"ZIP!" fall arresters. This enables multiple users to be connected to the system.

The GVS incorporates an intelligent cable vibration absorption element. This allows the lifeline to move with the tower when exposed to side forces such as wind and expansion/contraction due to temperature changes.

However, the system does restrain the worker to the line, and should the worker be required to move away from the installed system, shock absorbing lanyards must be taken up as well.



2. *Equipment*



Full Body Harness
(EN 361 & 358)



**Glider/ZIP! Fall
Arrester with rigid line**
(EN 353-1)



Shock Absorber
(EN 355)



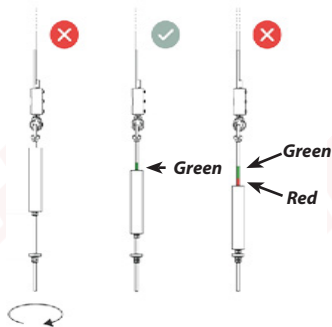
Workbelt
(EN 358)



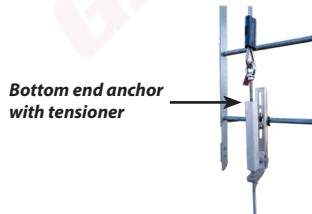
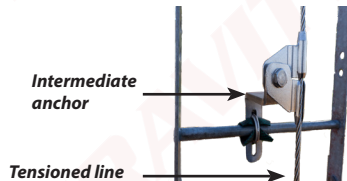
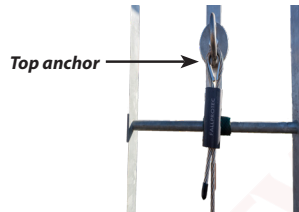
Helmet (EN 12492)

3. *Pre-Use Inspection of the GVS*

Always consult the Original Equipment Manufacturer (OEM) requirements. Before connecting to the GVS, the main elements of the system should be inspected and determined to be safe for use. If you are unable to perform a pre-use inspection on all of the main elements, ensure that the certificate plate indicates that the system is safe and then inspect the elements when you are in a position to do so.



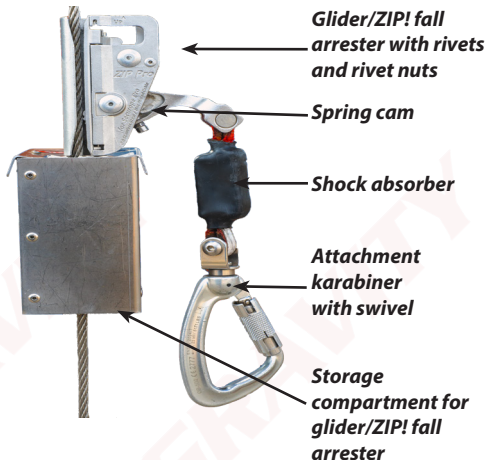
When tensioning the lifeline, take the colour code into account. Green indicates proper tension, red indicates too much tension and no colour indicates insufficient tension.



4. **Pre-Use Inspection of the Glider/ZIP!**

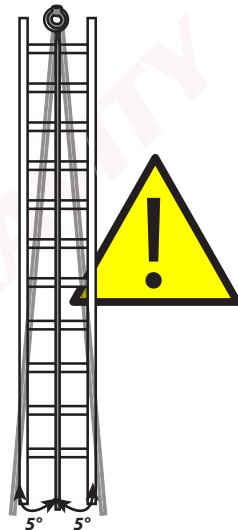
After the pre-use check of the system has been completed, the user needs to ensure that the glider/ZIP! fall arrester (EN 353-1) forms part of the GVS.

- The glider/ZIP! fall arrester needs to be compatible with the system.
- Do a function test on ground level by lifting the cam arm and letting go, the arm should spring back and the cam will engage on the system.
- Inspect the shock absorber for deployment and/or UV damage. Report it to the supervisor if it is damaged.
- Inspect all the other components of the glider/ZIP! fall arrester for any defects and/or damage before use.
- The attachment karabiner must freely swivel.



5. *Use and Limitations*

- The GVS must be clearly and correctly certified.
- The number of users allowed on the line is determined by the manufacturer and indicated on the certificate plate, but will not exceed more than four users for the system.
- In the event of multiple users on the system, it is best to maintain a minimum safe distance of 6 m while climbing. Should you need to work closer to each other, the top user needs to apply their work positioning lanyard (WPL) or remove themselves from the system by connecting an alternative safety to the structure and connecting the glider/ZIP! fall arrester to one of the ladder rungs.
- 5° maximum inclined angle as per OEM requirements.
- Only 50–120 kg users are allowed on the system.
- Only use the glider/ZIP! fall arrester.
- Keep your body close to the system to allow for better movement.
- 10 m maximum length between supporting anchors.
- 8 mm stainless steel wire rope.
- All components must be of compatible material.
- Do not use the system as a work positioning system.
- Do not stand or step on any of the components of the system.
- Do not use in inclement weather.
- If the line is found to be too slack, do not use it. Alert the site supervisor so that they may ensure that a competent person can inspect and adjust the line.



6. **Warnings**

- While at height, never remove the glider/ZIP! fall arrester from the line or from yourself before additional fall protection has been added.
- Do not manipulate the fall arrester directly.
- Do not use the glider/ZIP! fall arrester if its shock absorber is longer than 12 cm.
- Do not use if the line is slack.
- Do not ground a welding machine on a line.
- Never use the line outside of its limits.
- Do not use if the inspection is not current.
- Do not adjust, alter or tamper with the system in any way.
- Do not extend the glider/ZIP! fall arrester's shock absorber.



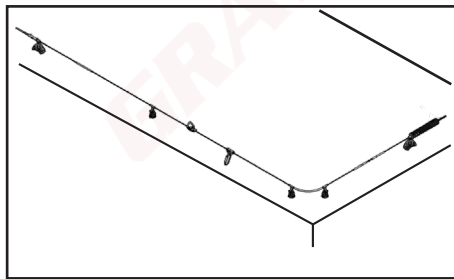
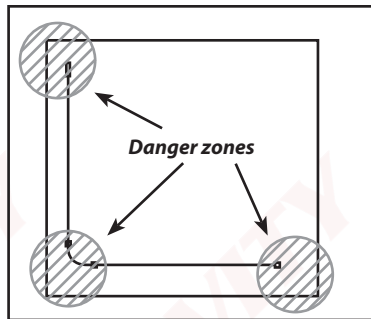
B. GRAVITY HORIZONTAL SYSTEM

The GHS is constructed with an 8 mm 7x7 stainless steel wire rope running through intermediate anchors. The GHS is connected to end anchors at both ends, usually with a tensioning device at one end. The horizontal line also includes a shock absorber which reduces the chance of injury in the case of a fall. The shock absorber also reduces the force on the system/structure.

1. *Advantages and Disadvantages*

This system offers some advantages over other fall protection systems. It reduces the fall risk that workers are exposed to. It improves work efficiency onsite as it allows for faster movement than would be possible with shock absorbing lanyards. And it gives access to areas where there are no other anchors available.

However, the system does restrain the worker to the line, and should the worker be required to move away from the installed system, shock absorbing lanyards must be taken up as well.



2. Equipment



Full Body Harness
(EN 361 & 358)



Glider
(EN 795-C)



Shock Absorber
(EN 355)



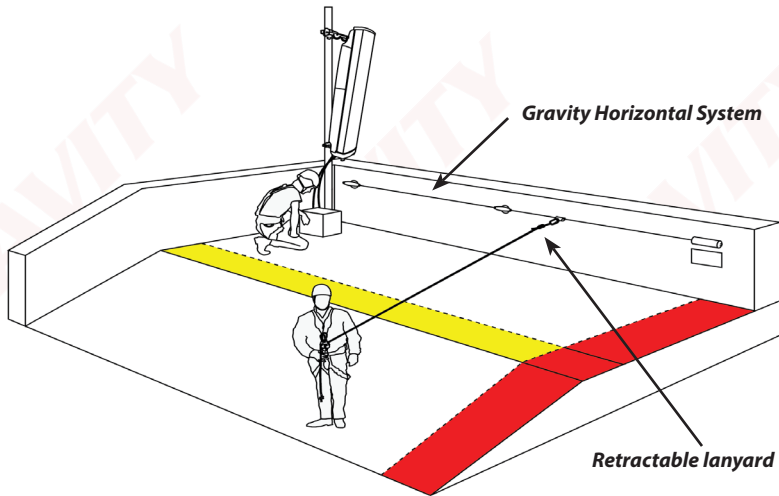
Helmet
(EN 12492)

****An EN 360 certified retractable lanyard may also be used.***



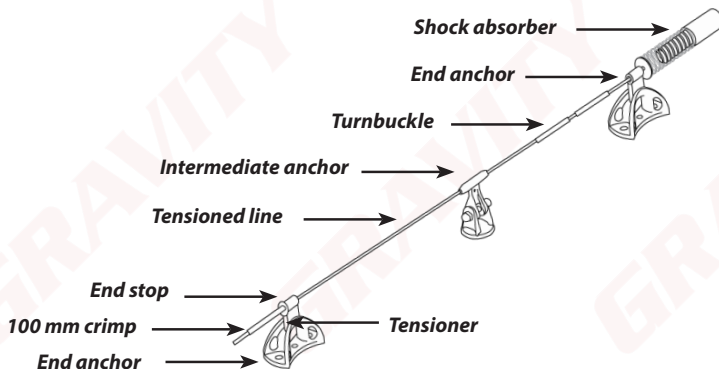
Connector
(EN 362)

3. *Use of a Retractable Lanyard on the GHS*

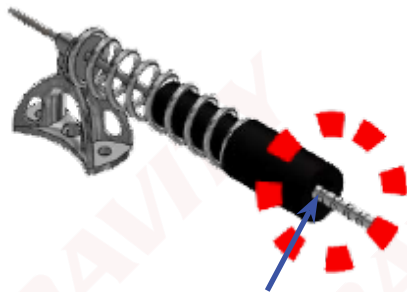


4. *Pre-Use Inspection of the GHS*

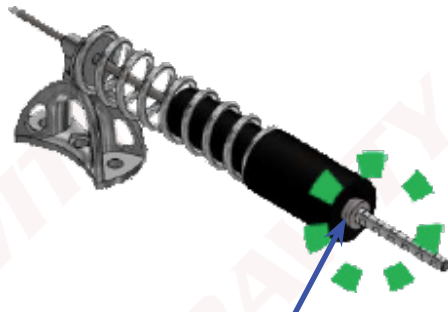
Always consult the OEM requirements. Before a user connects to the GHS, the following elements of the system must be inspected and determined to be safe for use.



5. *Fall Indicator*



Fall indicator activated = UNSAFE
Nut not visible



Fall indicator NOT activated = SAFE
Nut visible



6. *Use and Limitations*

- The GHS must be clearly and correctly certified.
- The amount of users allowed on the line is determined by the manufacturer and indicated on the certificate plate.
- Since the systems are always near edges or on rooftops, you should lower your centre of gravity by crouching when approaching or detaching from the system, to avoid potentially falling over the edge.
- All components must be of compatible material.
- Be careful not to trip over the lifeline.
- Always connect to the glider for easy movement over intermediate anchors.
- Tool bags of a maximum of 20 kg can be connected to the lifeline.
- 15° maximum inclined angle (as per EN 795:2012).
- Only 50–120 kg users are allowed on the system.
- 8 mm stainless steel wire rope.
- If the line is found to be too slack, do not use it. Alert the site supervisor so that they may ensure that a competent person inspects and adjusts the line.

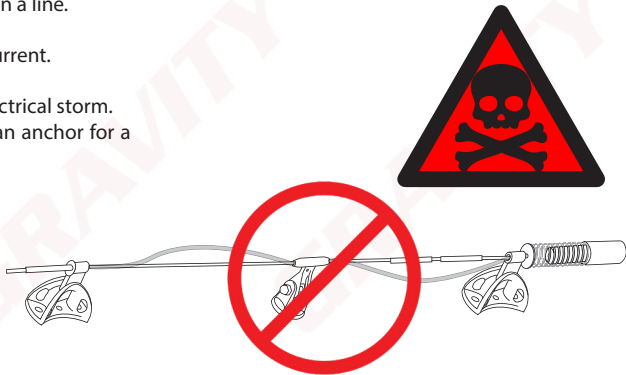


7. **Warnings**

- Never remove the captive glider from the line before additional fall protection has been added.
- Do not use the system if the fall indicator on the shock absorber has been activated.
- Do not use if the line is slack.
- Do not ground a welding machine on a line.
- Never use a line outside of its limits.
- Do not use if the inspection is not current.
- Do not use as rope access anchors.
- Do not use any lifeline during an electrical storm.
- Never use any overhead lifeline as an anchor for a lifting system.



**Damaged cable
(birdcage damage)**



C. THE WAY FORWARD

Congratulations on completing Gravity Training's Gravity Lifeline Systems course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map, or consult your facilitator/assessor.

Fall Arrest
Rigging
Rope Access
Included*



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graph LR
    BA[Basic Fall Arrest] --> AFA[Advanced Fall Arrest Rescue]
    BA --> FRU[Fallprotec Rail User]
    BA --> RAL1[Rope Access Level 1]
    BA --> RAL2[Rope Access Level 2]
    BA --> RAL3[Rope Access Level 3]
    BA --> CR[Climbing Equipment Inspection and Management]
    BA --> RR[Rope Rigging]
    BA --> ML[Mechanical Lifting]
    BA --> TE[Tower Erector]
    BA --> FAP[Fall Protection Plan Developer]
    BA --> RFA[Radio Frequency Awareness]
    BA --> PLP[Portable Ladder and Pole Climbing]
    BA --> PLU[Portable Ladder User]
    BA --> GVS[Gravity Vertical System and Gravity Horizontal System]
    GVS --> GVI[GVS Installer]
    RAL1 -- "+1000hrs" --> RAL2
    RAL2 -- "+1000hrs" --> RAL3
    RAL3 --> CR
    RR --> ML
    ML --> TE
    RFA --> FAP
    RFA --> PLP
    RFA --> PLU
    PLU --> OS[Optional + Solar]
  
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